

VisionInspect AI

Manufacturing Defect Detection & Quality Inspection System

Milestone 2 Report

PatchCore Anomaly Detection — Model Training & Evaluation

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Program: Infosys Springboard Internship 7.0

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Table of Contents

1. Work Summary
 - 1.1 Modules Completed
 - 1.2 Tools and Technologies Used
2. PatchCore Anomaly Detection
 - 2.1 Categories Trained
 - 2.2 Training Log Results
 - 2.3 Visual Inspection Outputs
3. Observations
4. Conclusion

1. Work Summary

1.1 Modules Completed

The following work was completed as part of Milestone 2, building on the dataset exploration and preprocessing pipeline established in Milestone 1:

- Trained the PatchCore anomaly detection model (Anomalib framework) on five MVTec AD product categories: Bottle, Cable, Capsule, Carpet, and Grid.
- Generated per-category test results including Image-level AUROC, Image-level F1 Score, Pixel-level AUROC, and Pixel-level F1 Score.
- Logged training completion details — elapsed training time and completion timestamp — for each category into a training log file (patchcore_training_log.json).
- Generated visual inspection outputs for each trained category, showing the input image, ground truth defect mask, anomaly heatmap, and predicted defect mask/boundary.

1.2 Tools and Technologies Used

Category	Technology
Programming Language	Python
Anomaly Detection Framework	Anomalib
Anomaly Detection Model	PatchCore
Dataset	MVTec AD

2. PatchCore Anomaly Detection

2.1 Categories Trained

PatchCore models were trained and tested independently for each of the following five product categories from the MVTec AD dataset (introduced during Milestone 1's dataset exploration): Bottle, Cable, Capsule, Carpet, and Grid.

2.2 Training Log Results

The table below summarizes the training completion log for each category, including the time taken to train and the resulting test metrics, as recorded in patchcore_training_log.json.

Category	Elapsed Time (s)	Image AUROC	Image F1	Pixel AUROC	Pixel F1
Bottle	158.7	1.0000	0.9920	0.9855	0.7263
Cable	185.0	0.9824	0.9674	0.9850	0.6385
Capsule	181.2	0.9908	0.9815	0.9901	0.5155
Carpet	257.1	0.9876	0.9773	0.9908	0.6064

Category	Elapsed Time (s)	Image AUROC	Image F1	Pixel AUROC	Pixel F1
Grid	224.3	0.9891	0.9649	0.9818	0.3854

Note: Timestamps recorded in the training log confirm each category's model was trained and tested sequentially — Bottle, Cable, Capsule, Carpet, and Grid.

2.3 Visual Inspection Outputs

For each trained category, the model output was visualized in four panels: the original input image, the ground truth defect mask, the image overlaid with the anomaly heatmap, and the image overlaid with the predicted defect mask/boundary.

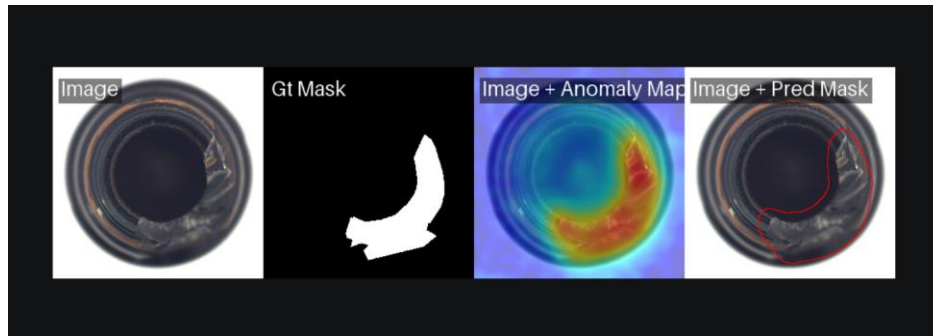


Fig 2.1: Bottle — Image, GT Mask, Anomaly Map, Predicted Mask

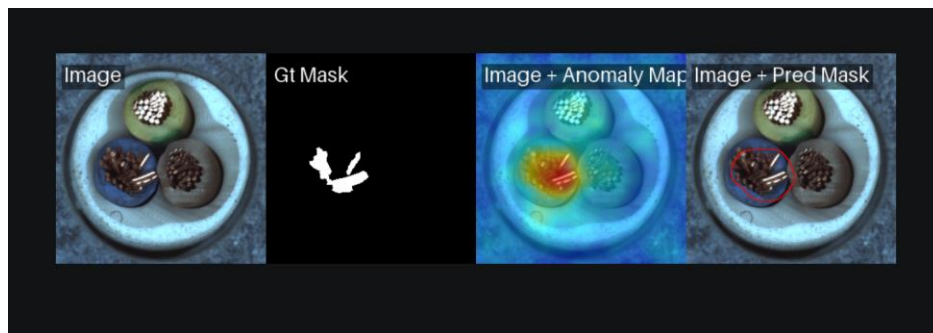


Fig 2.2: Cable — Image, GT Mask, Anomaly Map, Predicted Mask

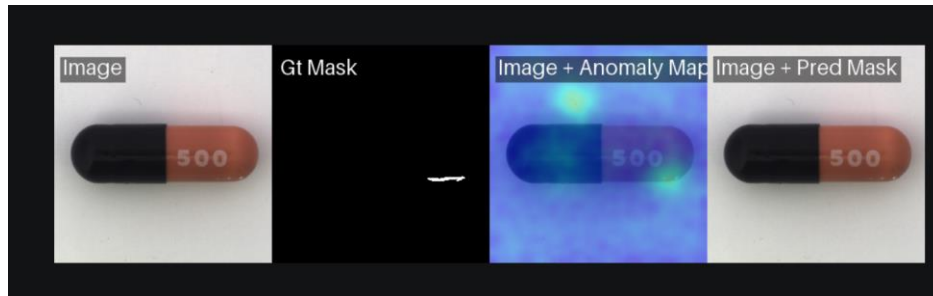


Fig 2.3: Capsule — Image, GT Mask, Anomaly Map, Predicted Mask

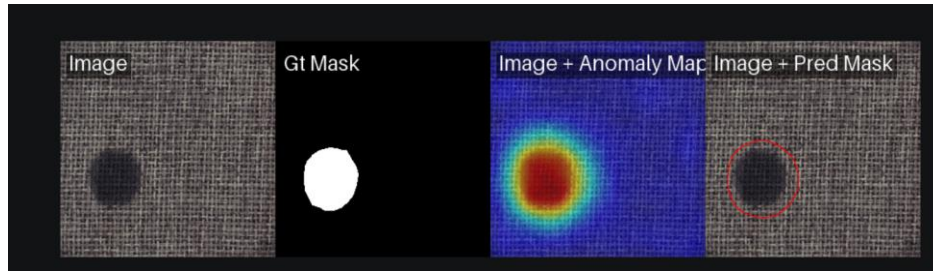


Fig 2.4: Carpet — Image, GT Mask, Anomaly Map, Predicted Mask

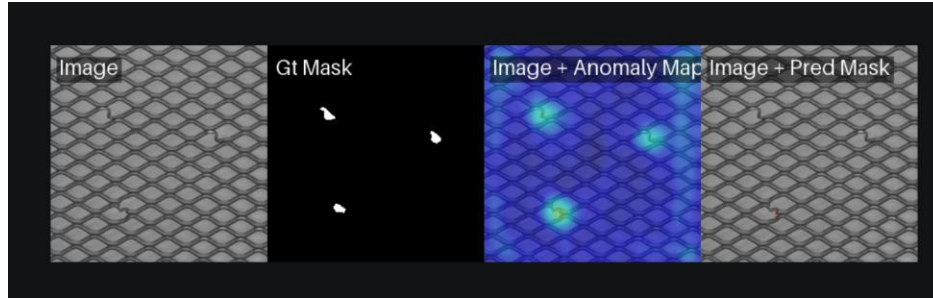


Fig 2.5: Grid — Image, GT Mask, Anomaly Map, Predicted Mask

3. Observations

- All five trained categories achieved Image-level AUROC above 0.98, with Bottle reaching a perfect 1.0000.
- Image-level F1 Scores remained high across all categories, ranging from 0.9649 (Grid) to 1.0000 (Bottle).
- Pixel-level AUROC was consistently high (above 0.98) for all categories, indicating accurate localization of defective regions.
- Pixel-level F1 Score varied more across categories (0.3854 to 0.7263), reflecting differences in defect size and shape across product types.
- Training time per category ranged from approximately 158.7 seconds (Bottle) to 257.1 seconds (Carpet).

4. Conclusion

Building on the dataset exploration and preprocessing pipeline completed in Milestone 1, Milestone 2 focused on training and testing the PatchCore anomaly detection model. Models were successfully trained on five MVTec AD product categories — Bottle, Cable, Capsule, Carpet, and Grid. Training logs and test metrics (Image AUROC, Image F1, Pixel AUROC, Pixel F1) were recorded for each category, and visual outputs were generated showing the input image, ground truth mask, anomaly heatmap, and predicted defect mask for each category.